

Chapter 04.

Classification Of Elements And Periodicity In Properties.

Periodic Laws.

Mendeleev's Periodic Table:

- The columns correspond to groups elements with same electrons in the outer-shell
↓
similar chemical properties.

- The rows correspond to periods elements who 2 electrons occupy the same number of shells in the ground state.

Modern Periodic Laws: The modern periodic law states that:

"The physical and chemical properties of the elements are periodic function of their atomic numbers."

Basis of Periodicity:

The basis of periodicity of the properties of elements is the repetition of similar electronic configuration of their atoms in the valence shell after regular intervals.

Example:

⇒ All alkali metals have this configuration: [noble gas] ns^1 .

⇒ All the halogens have this one. [noble gas] $ns^2 np^5$.

Periodic Table: It consists of

Periods and Groups.

Horizontal Rows
(we have 7 periods)

Vertical Columns

Remark:

Short periods: the first 3 periods contain 2, 8 and 8 elements respectively.

Long periods: the next 3 periods contain 18, 18 and 32 elements respectively.

The 7th period can have 32 elements but it is incomplete at present.

groups new system: it is sufficient to calculate the number of electrons in the valence shell to know the group's number.

groups old system: calculate the number of electrons in the valence shell, then identify the block then the group.

Example

New system:

Cu $\rightarrow 1s^2 2s^2 2p^6 3s^2 3p^6 3d^{10} 4s^1$
29
11 electrons
Valence shell

Hence Cu belongs to the group 11.

Old system:

K $\rightarrow 1s^2 2s^2 2p^6 3s^2 3p^6 3d^{10} 4s^2 4p^6$
36
8 electrons
Valence shell

Hence K belongs to the group (8A)

* If the valence shell contains either S or P block the element is in the group A. Otherwise it is in the group B.

Division of elements into s, p, d, and f blocks.

To know in which block the element exists

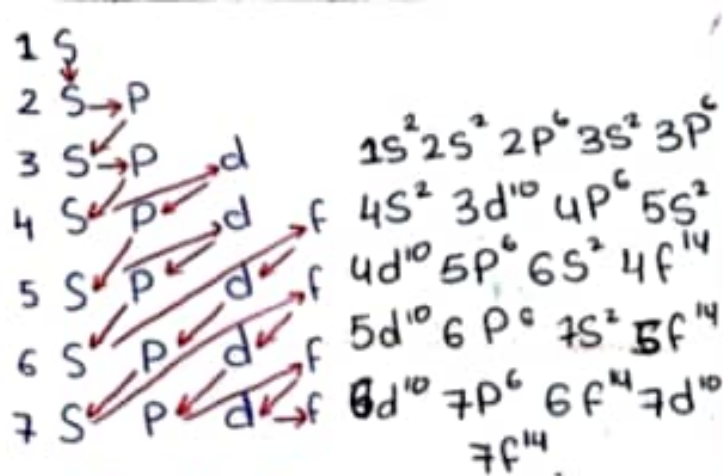
it is sufficient to know where its outermost electrons are located.

S-block: ns^{1-2}

P-block: $ns^2 np^{1-6}$

d-block: $(n-1)d^{1-10} ns^{0-2}$

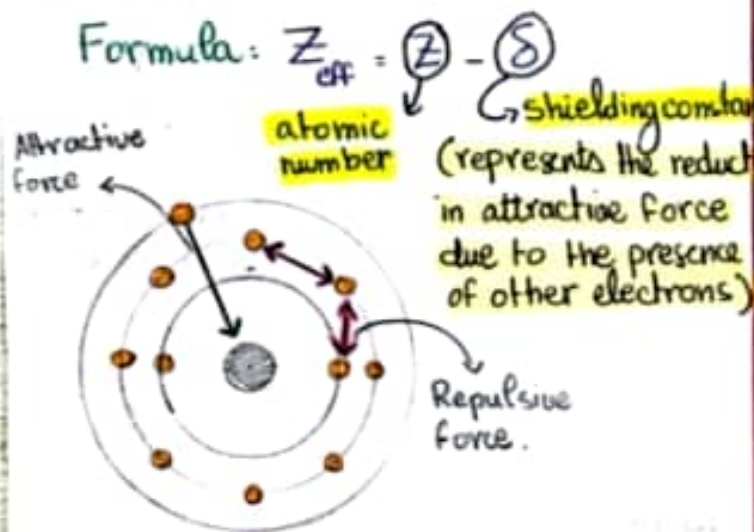
f-block: $(n-2)f^{1-14} (n-1)d^{0-2} ns^2$



* Periodic Properties,

Shielding and Effective Nuclear Charge

Formula: $Z_{\text{eff}} = Z - S$



Electron Type	Location	Contribution to Shielding
Any electron	to the right of electron of interest	0 (no contribution)
s and p electrons	Same group (same n)	0,35
	Same group, 1s orbital only	0,30
	Group immediately to the left (n-1)	0,85
	All other groups further left	1
d and f electrons	Same group (same n)	0,35
	All other groups further left	1

Atomic Radius: It is a measure of the effective size of an atom, defined as

the distance from the nucleus to the outermost boundary of the electron cloud.
 Across a period, Atomic radius decreases as you move from left to right.
 Down a group, Atomic radius increases as you move down.

! Note:

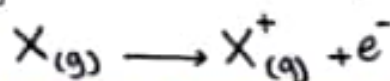
- The radius of a cation is smaller than that of its neutral atom.
- And the one of an anion is larger than that of its neutral atom.

Variation of size in an isoelectronic series

Definition of an Isoelectronic ions, they are those which have same number of electrons but differ from one another in the magnitude of the nuclear charge.

Due to increase in the nuclear charge, electrons are pulled more and more strongly and thus size decreases.

Ionization Energy: It is the energy required to remove an electron from a gaseous atom or ion.



it is expressed in terms of eV/atom or Kcal/mole or KJ/mole.

$$(1 \text{ eV/atom} = 23,06 \text{ Kcal/mole})$$

Across a period, ionization energy increases as you move from left to right.
 Within a group: ionization energy increases as you move from bottom to top.



Successive ionization energies: they refer to the amount of energy required to remove electrons from an atom or ion in a stepwise manner, starting

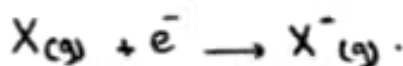
with the removal of the outer most electron.

Each successive ionization energy is larger than the previous one. This is because

• After each electron is removed, the remaining electrons experience a stronger effective nuclear charge due to reduced electron shielding.

• The ion becomes increasingly positive, making it harder to remove additional electrons.

Electron Affinity: It is the energy change when an electron is added to a neutral gaseous atom (Negative values $\rightarrow$ released, Positive values $\rightarrow$ absorption)

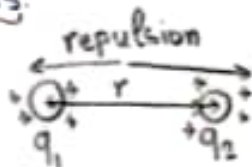


$r \searrow \rightarrow EA \nearrow$ same as ionization energy.

Successive Electron Affinity: like ionization enthalpies, the second and higher electron gain energies are also possible. However, the 2nd electron is added to an anion and the addition is opposed by coulombic repulsions.

the law:

$$F = k_e \frac{|q_1 \cdot q_2|}{r^2}$$



the energy has to be supplied to force the second electron into the anion.

Electronegativity: It is defined as the tendency of an atom to attract the shared pair of electrons towards itself when it forms a covalent bond with another atom in molecule.

It increases as IE and EA.

Different Scale of Electronegativity:

Mulliken Scale:

$$EN = \frac{IE + EA}{2}$$

Pauling Scale: Most widely used method.

$$X_A - X_B = 0.209 \sqrt{DE}$$

$$E_{AB} = \sqrt{E_{AA} \cdot E_{BB}}$$

Dissociation energies

The Relation between the two Scales:

$$X_{\text{Pauling}} = \frac{X_{\text{Mulliken}}}{2.8}$$

Electronegativity has no unit.

Metallic and Non-metallic Property:

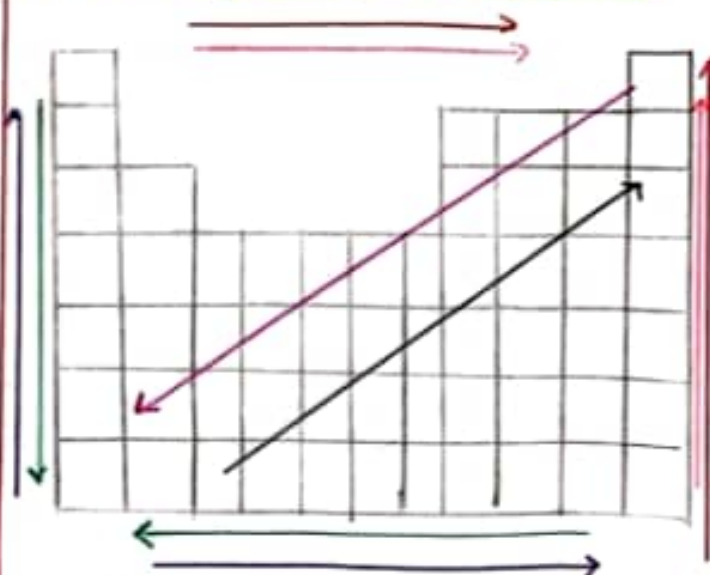
Metals generally tend to lose electrons easily, forming cations. While non-metals tend to gain electrons, forming anions.

(a) Period: From left to right

$EN \nearrow \rightarrow$ Metallic character $\searrow \rightarrow$ Non-M.C $\nearrow$

(b) Group: From top to bottom.

$EN \searrow \rightarrow$ M.C $\nearrow \rightarrow$ Non-M.C $\searrow$



$\rightarrow$ = IE.

$\rightarrow$ = EN.

$\rightarrow$ = M.C.

$\rightarrow$ = Non-M.C.

$\rightarrow$ = r (atomic radius)

$\rightarrow$ = EA.